Spring 2026

DATA 505, Applied Machine Learning

FITTING FIRE WITH FIRE (FEATURES):

MACHINE LEARNING TO PARSE HUMAN FROM NATURAL CAUSES OF FOREST FIRES

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BACKGROUND

WILDFIRE
IN OREGON

Oregon is one of the most wildfire prone states in the US!

In 2024, a record-breaking **1.9 MILLION ACERS** were burned across Oregon.

SEVERITY OF FIRE SEASON depends on many factors, such as:

- Drought
- Snowpack levels
- Weather conditions

WHAT IGNITES WILDFIRE?

CAUSAL TREND

- Naturally caused (lightning!): early season
- Human-caused: late season

THE POWER OF MACHINE LEARNING

By identifying key predictors, Oregon can be better prepared to prevent and fight wildfires.

RESEARCH QUESTION

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“In Oregon, how do environmental and human-related factors differently predict human-caused versus lightning-caused wildfires?”

GOALS

1. How does this model perform at predicting wildfires generally?
2. How does this model perform at predicting lightning vs. human wildfires?

BUILDING ON MATHUR ET AL. (2025)

WHAT WE BORROWED

- Data
- Features
- Binary fire predictions

WHERE WE DIFFERED

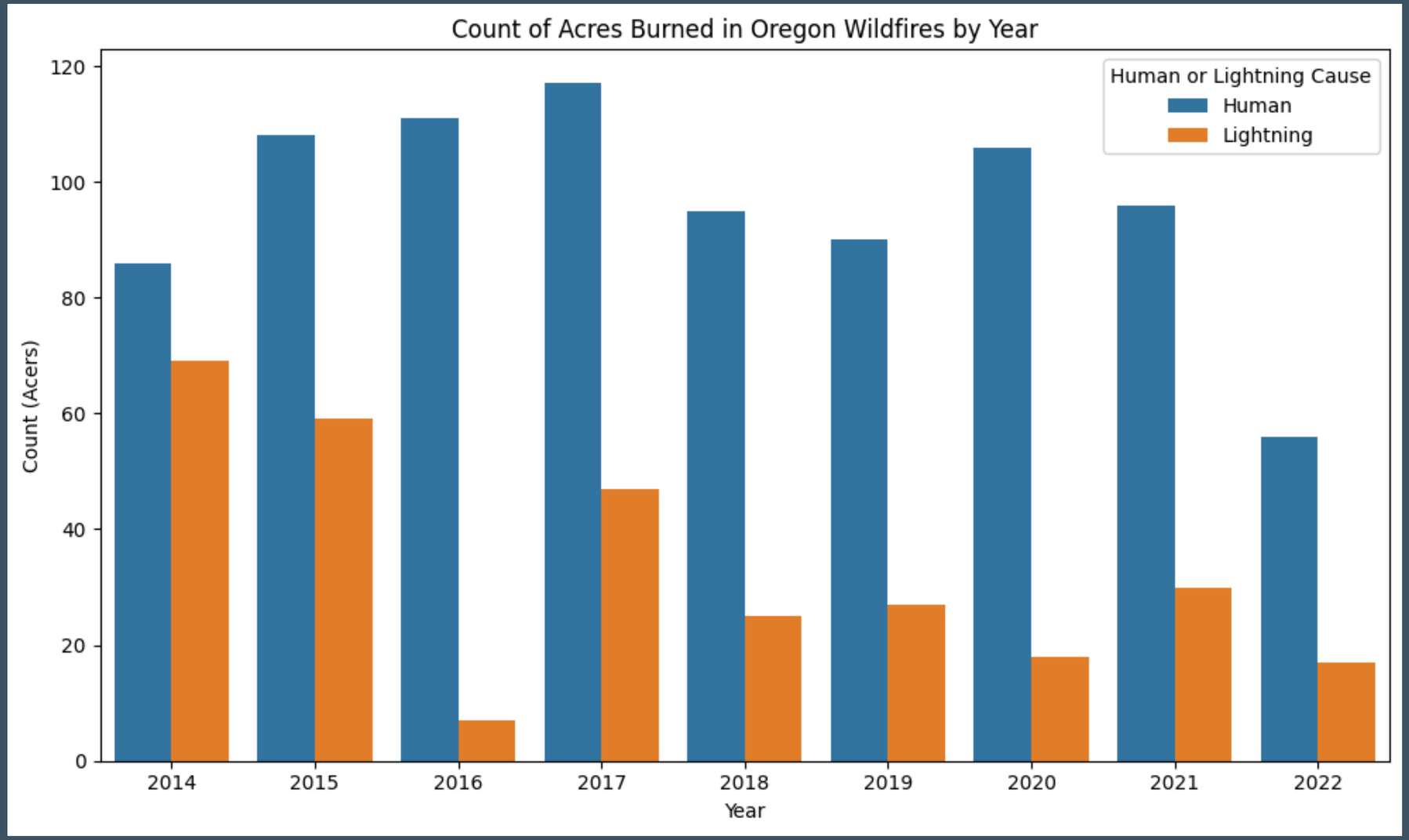
- Added cause classification
- Used simpler models
- Limited scope to Oregon
- More engineered features

HOW WE COMPARE

- Their model: 73.1% accuracy, 0.70 recall
- Our model: similar accuracy, similar recall

COMPARING GROUPS

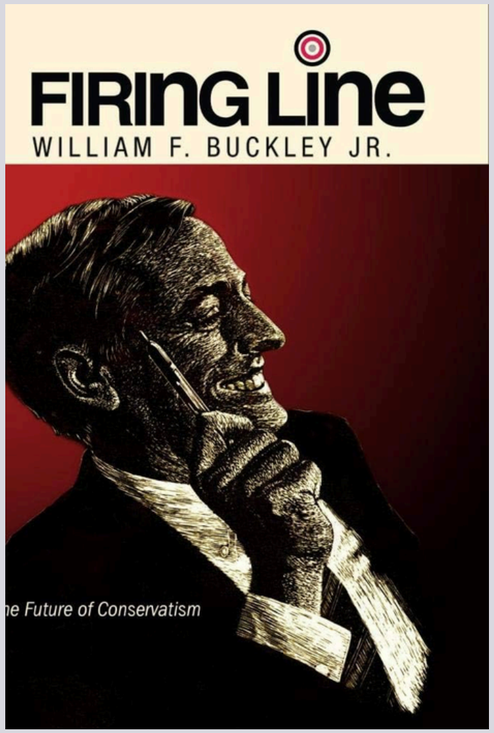
HUMAN VS. LIGHTNING: ACERS BURNED



MODELS

MODEL COMPOSITION

Before creating our ensemble model, we first trained and tested **six models** (e.g., three KNN models, a decision tree, random forest, and gradient boosting). We chose one model of each type based on the best recall score. **Three models was ideal for hard voting** (e.g., an odd number to prevent tied votes)



CONSERVATIVE MODELING

To prevent over fitting from GridSearch parameter tuning and class imbalance, we employed **hard voting** (i.e., model agreement vs. probability) and **class weights**, respectively. We used **recall scoring** in training to penalize false negatives.

Fire Feature Models

STATE

Model	Accuracy [%]	Precision	Recall	F1
Random Forest	**79**	**78**	**79**	**78**
K-Nearest Neighbors	74	73	74	73
Gradient Boosting	72	73	72	72
CNN-LSTM (target)	**73**	**71**	**70**	**70**

* Models sorted by recall in descending order

** Target model from Mathur et al. (2025) was trained on U.S. Wildfire Data

RECALL

We chose recall as our performance indicator to **emphasize the least false negative** (i.e., predicting no fire when a fire actually occurred)

$$Recall = \frac{TP}{TP + FN}$$

HTTPS://WWW.AMAZON.COM/FIRING-WILLIAM-BUCKLEY-FUTURE-CONSERVATISM/DP/B001E52U2G

DATA ON A WHOLE OTHER SCALE

srad	6078.638128
sph	0.000004

Before training our models, we transformed our data using a pipeline. For example, models on completely different scales were processed using a logarithm to prevent high-variance data from dictating results.

HTTPS://WWW.GETTYIMAGES.COM/PHOTOS/BIG-DOG-LOOKING-DOWN

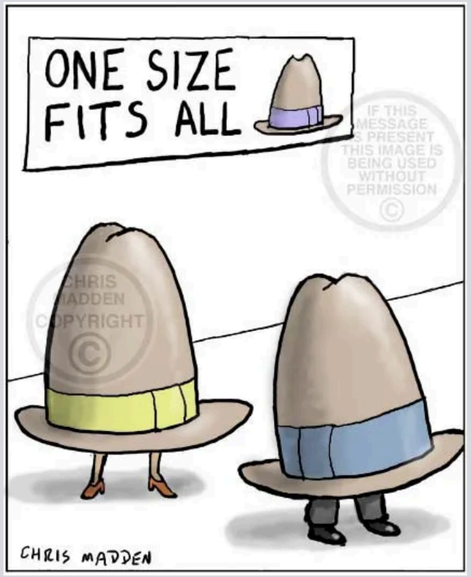
HTTPS://LEVELUP.GITCONNECTED.COM/DATA-PREPROCESSING-IN-MACHINE-LEARNING-D3A48D2AAB2B

OUR APPROACH

We chose a one-size-fits-all approach to data transformation. Models like KNN require centering (e.g., distance from center) while random forest is agnostic to centering but sensitive to class imbalance. For simplicity, we chose to fit the ensemble features with **one pipeline.**

PIPELINE

For values like solar radiation (srad) and specific humidity (sph), we used **log transformer**. For index values (e.g., 0-100), we used **MinMaxScaler** to normalize to [0,1]. Otherwise, we used **standard scaler** to center.



https://www.chrismadden.co.uk/cartoon-gallery/absurdist-fashion-cartoon-one-size-fits-all/

TRAINING

The emsemble model was trainged on three data sets: **Most Recent Year (MRY)** which included data from **2014 to 2021**, and **Holiday Farm** and **Chaos Rough-Patch**, both of which contained **data up to 30 days before the fires were reported**.

TESTING

HTTPS://WWW.EDUTOPIA.ORG/BLOG/PREPARING-STUDENTS-STATE-STANDARDIZED-TESTS

The MRY data was tested using all data from 2022 to see how well data from previous years predicted the more recent year in our data. The Holiday Farm (2020) and Chaos Rough-Patch (2021) fire were single cases we used for **fire prediction** and **human vs. lightning predictions**.

DATA SPLIT

Training on all data up to the **MRY**, to predict MRY was our **analog** to the **Mather et al. (2025)** study.

We tested on **single cases** of **Oregon's largest fires** in our data range (e.g., the **Holiday Farm** fire burned **173,339 acres**). This was to achieve parity with Mather et al. (2025) whose model predicted the **Palisades Fire** (**23, 448 acres**).

HTTPS://WWW.TRYPLAYGROUND.COM/BLOG/10-TIPS-FOR-MANAGING-A-CLASSROOM-OF-YOUNG-CHILDREN

HTTPS://KVAL.COM/NEWS/LOCAL/LIKE-A-BOMB-HIGH-WINDS-FUEL-DESTRUCTIVE-PACIFIC-NORTHWEST-BLAZES

RESULTS

RESEARCH QUESTION GOAL #1

PREDICTING FIRE EVENT

Testing Most Recent Year

Fire Feature Models

STATE / Model	Accuracy [%]	Precision	Recall	F1
Random Forest	**79**	**78**	**79**	**78**
K-Nearest Neighbors	74	73	74	73
Gradient Boosting	72	73	72	72
Ensemble (Weighted)	**71**	**71**	**71**	**71**
CNN-LSTM (target)	**73**	**71**	**70**	**70**
Ensemble	**70**	**70**	**70**	**70**

* Models sorted by recall in descending order

** Target model from Mathur et al. (2025) was trained on U.S. Wildfire Data

Ensemble is very similar to target model!
Increased confidence in extending research to compare human vs. lightning.

With respect to **recall**, our model **underperformed** the initial models, but **outperformed** the target model in Marthur et al. (2025).

Important Reminder!

Initial models do not include penalties that our ensemble DOES!

(class weights, recall scoring, hard voting)

RESEARCH QUESTION GOAL #1

PREDICTING FIRE EVENT

TESTING SPECIFIC FIRES

using data from 2014 until 30 days prior

Holiday Farm Fire
173,393 acres
Natural Caused

ACCURATELY PREDICTED LIGHTNING CAUSED

Chaos Rough-Patch
50,409
Human Caused

FAILED TO PREDICT HUMAN CAUSED

Human wildfires are very unpredictable events!

RESEARCH QUESTION GOAL #2

PREDICTING HUMAN OR LIGHTNING

Testing Most Recent Year

Lightning / Human Feature Models

STATE	Accuracy [%]	Precision	Recall	F1
Model				
Random Forest	**95**	**95**	**95**	**95**
Gradient Boosting	93	93	93	93
K-Nearest Neighbors	92	92	92	92
Ensemble	**92**	**92**	**92**	**91**
Ensemble (Weighted)	**85**	**85**	**85**	**85**

* Models sorted by recall in descending order

With respect to **recall**, our model **underperformed** the initial models, but **values remain high.**

Important Reminder!

Initial models did not include penalties that our ensemble DOES!

(class weights, recall scoring, hard voting)

RESEARCH QUESTION GOAL #1

PREDICTING FIRE EVENT

TESTING SPECIFIC FIRES

using data from 30 days prior

Holiday Farm Fire

173,393 acres

Natural Caused

ACCURATELY PREDICTED LIGHTNING CAUSED

Chaos Rough-Patch

50,409

Human Caused

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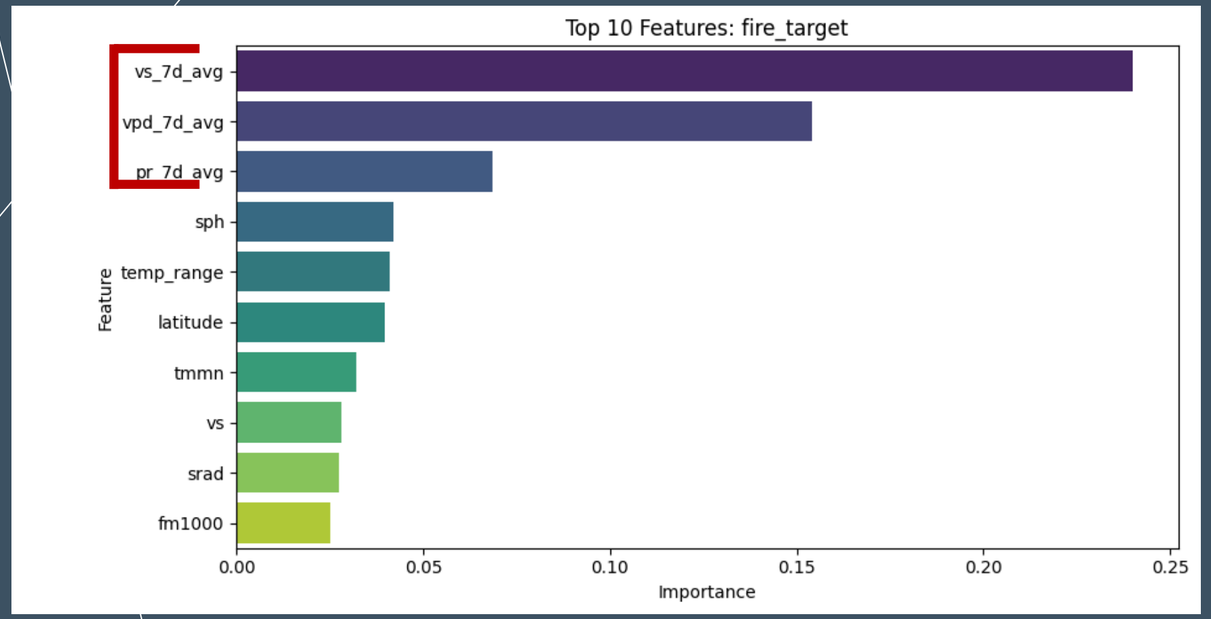
ENSEMBLE WEIGHTED

WHAT ARE THE TOP PREDICTORS?

WILDFIRE EVENT

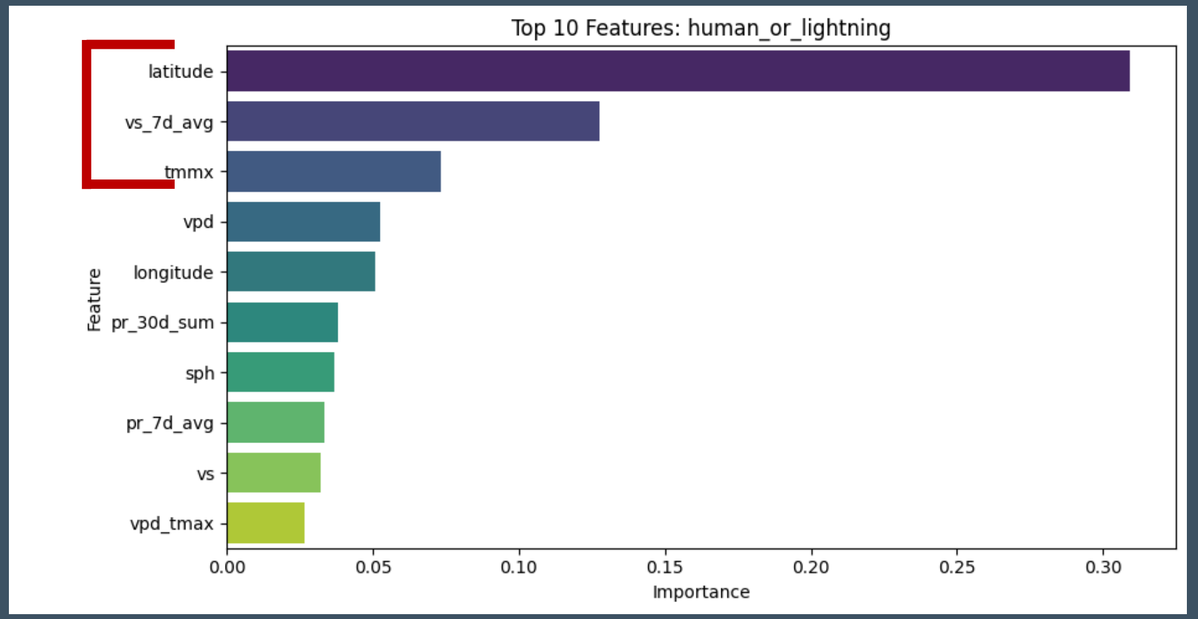


TOP PREDICTORS:

Weather features!

1. Wind speed
2. Vapor pressure deficit (dryness)
3. Precipiation

HUMAN OR LIGHTNING



TOP PREDICTORS:

Geographic *then* weather features

1. Latitude
2. Wind speed
3. Heat stress

CONCLUSION

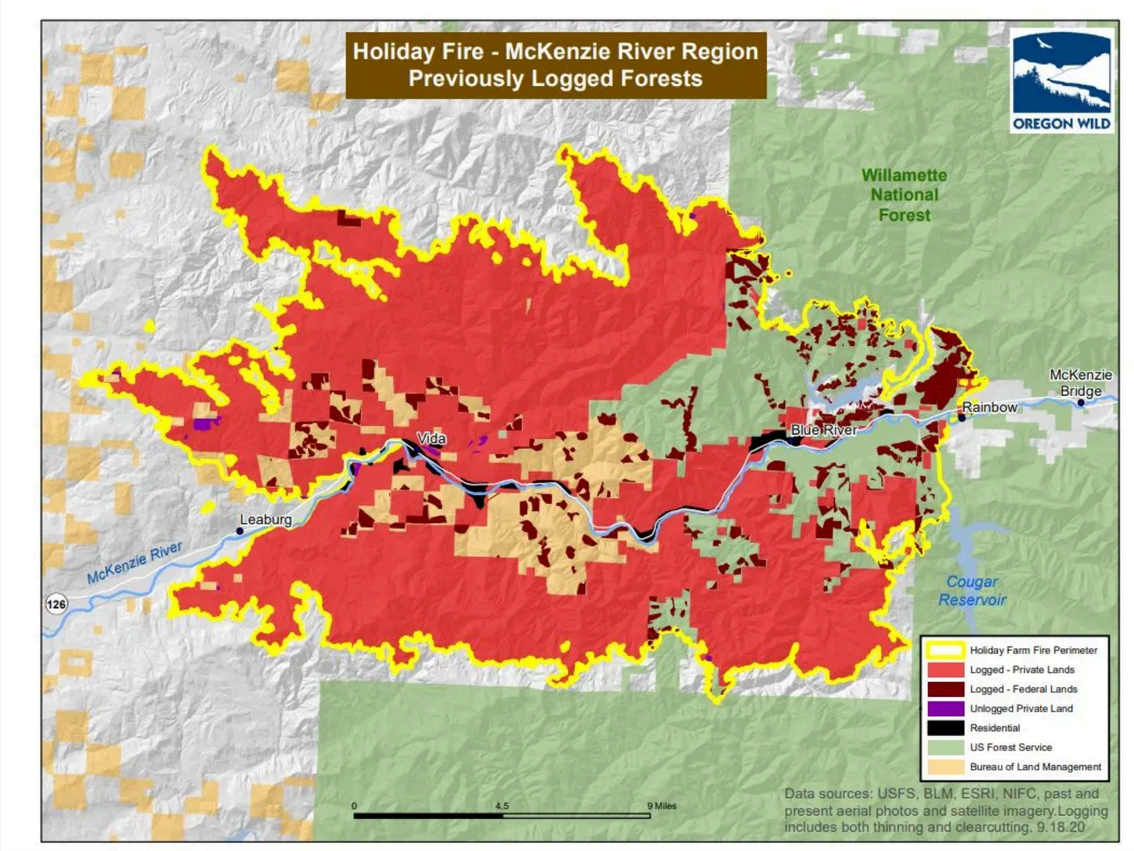
CONCLUSION

Strong prediction of wildfire occurrence

TOP PREDICTORS :

1. Wind Speed
2. Vapor Pressure Deficit
3. Precipitation

+ Geological variables (lat/long) for human vs. lightning



Environmental predictors were effective for wildfire occurrence prediction.

LIMITATIONS

Human-caused ignitions may be harder to predict because they arise from less predictable factors.

FUTURE DIRECTIONS

- **STAKEHOLDER INTEGRATION** – DIFFERENT USERS DEFINE GOOD MODEL PERFORMANCE DIFFERENTLY
- **RECALL AS A DESIGN CHOICE** – MINIMIZING FALSE NEGATIVES HELPS PREVENT CATASTROPHE, BUT CONTEXT MATTERS
- **FEATURE EXPANSION** – LAND USE, ROAD PROXIMITY, AND HUMAN ACTIVITY PATTERNS WOULD IMPROVE CAUSE CLASSIFICATION; ACREAGE BURNED WOULD ENABLE SEVERITY PREDICTION
- **MODEL COMPLEXITY** – A MULTI-CLASS CAUSE MODEL WOULD BE MORE ACTIONABLE THAN THE CURRENT HUMAN VS. LIGHTNING BINARY

THANK YOU!

SOURCES

Davis, Erin & Harris, Emily. July 21, 2023. "People cause about half of Oregon wildfires." Axious Portland. https://www.axios.com/local/portland/2023/07/21/people-cause-half-of-wildfires-in-oregon.

Western Fire Cheifs Association. Feb 22, 2023. "Oregon Fire Season: In-Depth Guide". https://wfca.com/wildfire-articles/oregon-fire-season/.

Plante, Aimee. Oct 28, 2024. "Oregon's record-breaking wildfire season ends with more than 1.9 million acres burned". https://www.koin.com/news/wildfires/oregon-2024-fire-season-over-10282024/#:~:text=This%20season%2C%20which%20took%20off,strained%20resources%20and%20exhausted%20firefighters.

Mathur et al. 2025. "Spatiotemporal Wildfire Prediction and Reinforcement Learning for Helitack Suppression". IEEE. https://arxiv.org/pdf/2601.14238.